SOFTWARE ENGINEERING LAB 3

Software Architecture & Component Design

Corporate Mental Wellness & Counseling Platform

STUDENT DETAILS	INFORMATION
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1. Purpose and System Context

This document defines the software architecture for the Corporate Mental Wellness & Counseling Platform. The system provides employees with an anonymous route to stress assessment, confidential therapist appointment booking and personal wellness tracking. Licensed therapists use the platform to view assigned appointments through employee pseudo-IDs rather than corporate identity fields.

The design is derived from the five functional requirements (FR-001 to FR-005) and two non-functional requirements (NFR-001 and NFR-002) recorded in the team's Lab 1 requirements specification. Lab 2 groups those functions into Employee Wellness & Assessment, Confidential Counseling & Appointment Booking, and Therapist Appointment Management.

1.1 Stakeholders and scope

Stakeholder	System interaction	Information boundary
Corporate Employee	Uses pseudo-ID account, completes stress assessment, books sessions, records wellness indicators.	Can access own assessment, booking and wellness information.
Licensed Therapist	Views assigned schedule and appointments, identifies clients using pseudo-IDs.	Must not see employee corporate email, employee ID or employer identity.
Platform operator / administrator	Maintains service availability, access controls and system configuration.	Operational access is controlled and audited; not a business actor in Lab 1 scope.

1.2 Design principles

- Confidentiality by design: therapist-facing interfaces expose pseudo-IDs, not employee identity.
- Separation of concerns: routing, authentication, anonymization, assessment, booking, wellness tracking and persistence are distinct responsibilities.
- Least privilege: role-based authorization constrains employee and therapist operations.
- Data integrity: booking availability is checked transactionally to prevent duplicate slot confirmation.
- Traceability: each major component maps to one or more functional or non-functional requirements.

2. Requirements Baseline

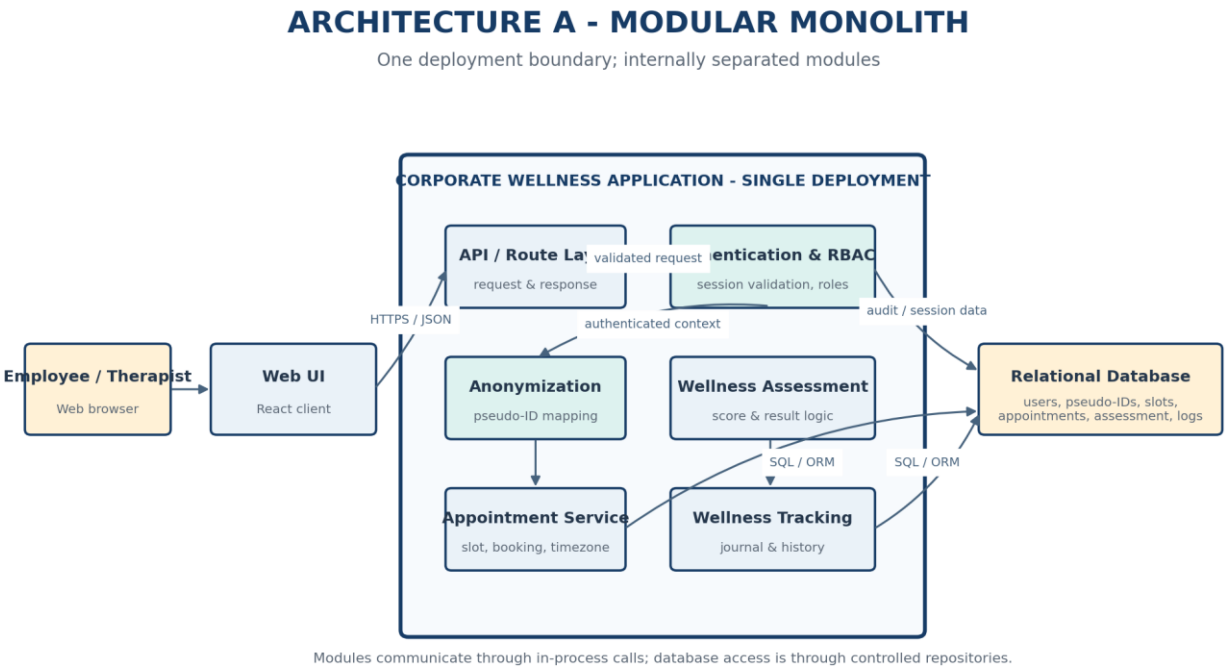
The table below summarizes the agreed requirements from Lab 1. Requirement IDs are retained to make architecture and component decisions traceable.

ID	Requirement summary	Priority
FR-001	Generate an anonymized pseudo-ID so employees can book counseling without exposing corporate email or employee ID to the therapist.	High
FR-002	Allow employees to complete a stress-level assessment and view the result.	High
FR-003	Show available appointment slots and book confidential counseling sessions with licensed therapists.	High
FR-004	Allow therapists to view assigned appointments by employee pseudo-ID without corporate identity access.	Medium
FR-005	Allow employees to record and track personal wellness indicators over time.	Medium
NFR-001	Handle time-zone conversion accurately across global offices; retain suitable performance under peak load.	High
NFR-002	Protect employee counseling and wellness data from unauthorized access and disclosure.	High

3. Architecture Alternatives

3.1 Architecture A - Modular Monolith

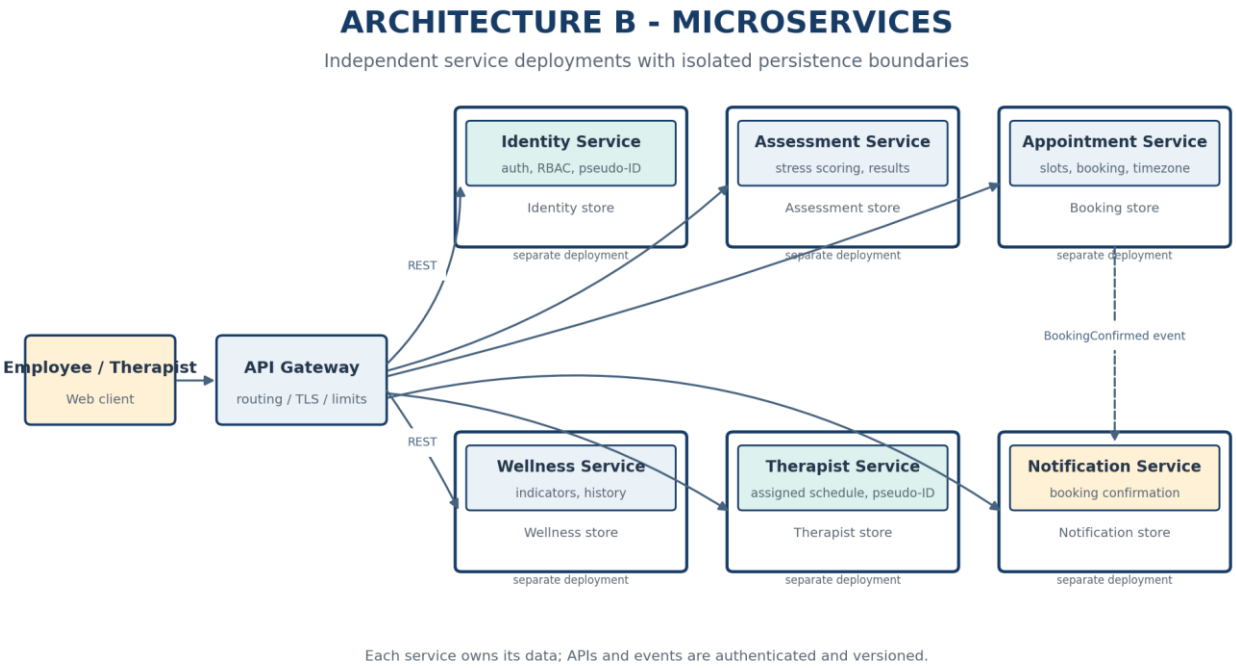
In the modular monolith, the web client communicates with one deployed backend application. The backend is internally divided into route/API, authentication and role authorization, anonymization, stress assessment, appointment management, wellness tracking, therapist workspace and repository/data-access modules. The modules call one another in-process and share a controlled relational database.



- The client sends an HTTPS request to the API/route layer.
- The authentication and RBAC module validates the session and role before protected operations continue.
- The route layer delegates to the relevant business module: anonymization, assessment, appointment booking, wellness tracking or therapist workspace.
- Business modules use the repository/data-access layer to read or persist records in the relational database.
- The API layer returns a response to the client. Therapist-specific queries return pseudo-ID references only.

3.2 Architecture B - Microservices

In the microservices alternative, the API gateway routes requests to independently deployable Identity, Assessment, Appointment, Wellness, Therapist and Notification services. Each service owns its data store or schema boundary and communicates through versioned REST APIs and asynchronous events where appropriate. Identity and pseudo-ID mapping are isolated from therapist appointment views.



- The web client sends requests to the API gateway, which applies request routing and edge-level controls.
- The gateway forwards each operation to its owning service. Identity operations are handled by the Identity Service; booking requests are handled by Appointment Service; wellness and assessment operations go to their respective services.
- Services call one another through authenticated, versioned APIs or publish events such as BookingConfirmed.
- Each service persists data within its own database boundary.
- The gateway aggregates or forwards the response to the requesting client.

4. Architecture Comparison

Decision criterion	Modular monolith	Microservices
Deployment	One backend application is packaged and released as a unit.	Each service can be deployed and rolled back independently.
Communication	In-process method/module calls; low communication overhead.	REST/API and event-based network communication; adds latency and network failure modes.
Scaling	All backend modules scale together, even if only booking is busy.	Appointment, assessment or wellness services can scale independently.
Failure impact	A process-level fault may affect multiple platform functions.	Service isolation can limit impact, although dependencies still need timeouts and fallbacks.
Data management	Modules share one controlled relational database and transaction boundary.	Data ownership is distributed; cross-service consistency requires APIs/events and compensating workflows.
Operational complexity	Lower deployment, monitoring and debugging overhead.	Higher infrastructure, observability, API versioning and incident-management overhead.
Security boundary	RBAC and module-level authorization enforced inside one backend.	Each service needs service authentication, authorization and secure inter-service communication.
Best fit	Student prototype, small team, moderate traffic and tightly related workflows.	Large scale, multiple independent teams, different scaling profiles or independent release cadence.

5. Selected Architecture and Justification

Selected design: Modular monolithic backend with clearly separated modules and a relational database.

The modular monolith is selected for the current student project because the platform is being developed by a small team, its workflows share a closely related domain and data model, and there is no evidence in the current requirements of traffic volume or independent service ownership that

would justify distributed deployment. It reduces network failure modes and makes end-to-end testing of authentication, anonymization and booking transactions easier.

The choice does not imply that confidentiality is optional. Security boundaries are enforced in the modular monolith through authentication, role-based authorization, explicit therapist data-transfer objects that contain pseudo-IDs only, parameterized database access, and audit logging. The identity-to-pseudo-ID mapping is isolated behind the anonymization component and is never returned in therapist-facing responses.

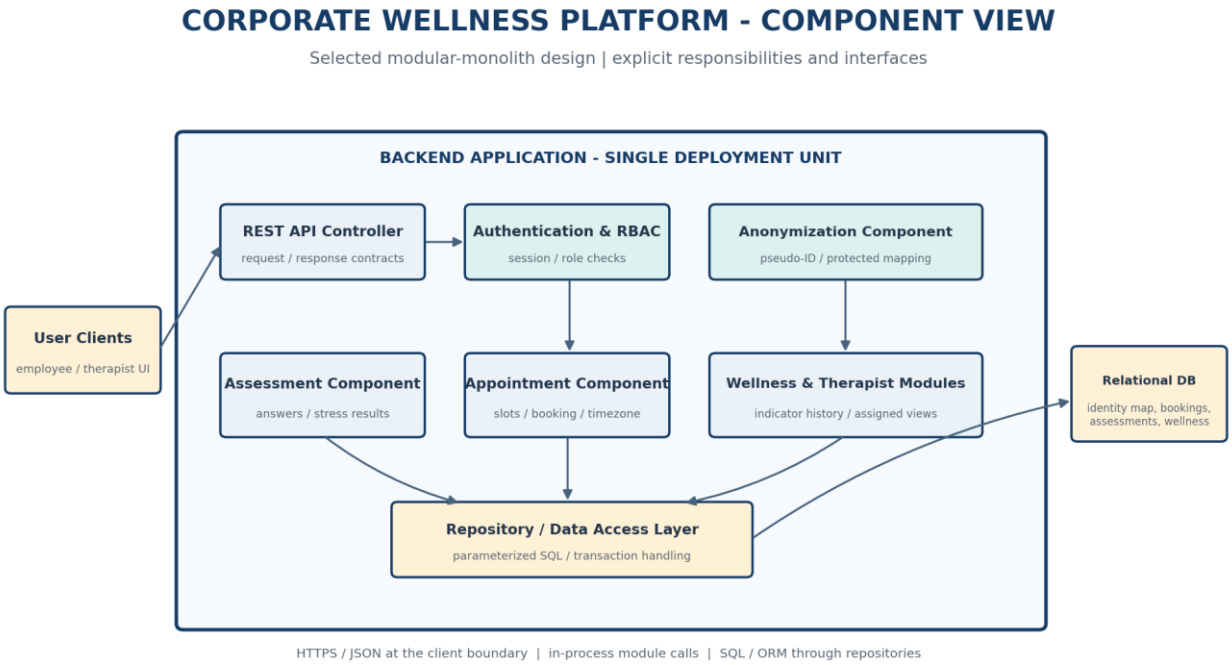
Evolution path: if the system later experiences substantial booking load, requires independent release cycles, or gains separate service-owning teams, the Appointment Service or Notification Service can be extracted first. The modular interfaces and repository boundaries are intended to reduce that future migration effort.

6. Component Design

Component	Responsibility	Key interfaces / operations	Requirements
Web UI	Employee and therapist screens; collects input and renders results.	HTTPS JSON requests; employee dashboard; therapist dashboard.	FR-002 to FR-005
API / Route Layer	Defines request/response contracts, routes calls and maps errors to HTTP responses.	POST assessment; GET slots; POST booking; GET therapist appointments; wellness CRUD.	FR-002 to FR-005
Authentication & RBAC	Validates sessions and applies employee/therapist role permissions.	authenticate(token); authorize(role, resource, action).	NFR-002
Anonymization	Creates pseudo-IDs and isolates identity-to-pseudo-ID mapping.	createPseudoId(employee); resolveMapping(internal-only).	FR-001, FR-004, NFR-002
Assessment Module	Validates assessment answers, calculates stress result and stores result.	submitAssessment(answers); getAssessmentResult(employeePseudoId).	FR-002
Appointment Module	Lists available time slots, checks conflicts, confirms bookings and handles time zones.	getAvailableSlots(therapist, range, timezone); bookSlot(pseudoId, slotId).	FR-003, NFR-001, NFR-002
Wellness Tracking Module	Records and retrieves employee wellness indicators over time.	addIndicator(pseudoId, metric, value, timestamp); listIndicators(pseudoId).	FR-005, NFR-002
Therapist Workspace Module	Provides therapist schedule and assigned appointment details using pseudo-ID only.	getAssignedAppointments(therapistId); getClientPseudoProfile(pseudoId).	FR-004, NFR-002
Repository / Data Access	Encapsulates SQL, parameterized queries and transaction handling.	EmployeeRepository; BookingRepository; AssessmentRepository; WellnessRepository.	FR-001 to FR-005, NFR-002
Relational Database	Persists accounts, protected identity mapping, pseudo-IDs, slots, appointments, assessments, wellness indicators and audit events.	Transactional CRUD; unique constraints on booking slot allocation.	FR-001 to FR-005, NFR-001, NFR-002

6.1 UML Component Diagram

The UML component view below summarizes the selected modular-monolith design, its interfaces and the persistence boundary. The editable diagrams.net source is included alongside this report.



7. Component Interaction and Data Flows

7.1 Anonymous employee onboarding / pseudo-ID

- Employee submits the onboarding request through the Web UI.
- The API layer validates the request and delegates authentication/session creation to the Authentication & RBAC module.
- The Anonymization component generates a unique pseudo-ID and stores the protected mapping in the restricted identity repository.
- The employee-facing response returns the pseudo-ID; corporate identity fields are not included in therapist-facing appointment records.

7.2 Stress assessment

- The employee submits assessment responses through the API.
- The Assessment Module validates the answers, calculates the result and associates the assessment with the employee account/pseudo-ID.
- The result is persisted and returned to the employee dashboard.

7.3 Confidential appointment booking

- The employee requests available therapist slots and provides the relevant time zone.
- The Appointment Module converts and displays slot times in the requested time zone using a canonical stored time representation (UTC).
- On selection, the module rechecks slot availability within a database transaction and applies a uniqueness constraint to prevent duplicate booking.
- The booking is persisted with the pseudo-ID. The therapist view receives appointment time and pseudo-ID without employee corporate identity.
- The API returns booking confirmation to the employee.

7.4 Therapist appointment view

- A therapist authenticates and requests their assigned appointments.
- RBAC confirms therapist role and assignment scope.
- The Therapist Workspace obtains appointment details and uses a therapist-safe response model containing pseudo-ID only.
- The UI displays assigned sessions; the identity mapping remains inaccessible to therapist-facing components.

7.5 Wellness tracking

- The employee submits a wellness indicator.
- The Wellness Tracking Module validates the metric and value, records the timestamp and persists the record.
- The employee can retrieve the indicator history for personal progress review.

8. Interface Contracts

Interface	Example contract	Important controls
Assessment submission	POST /api/assessments { answers: [...] }	Authenticated employee; validated answer schema; response only to owner.
Available slots	GET /api/therapists/{id}/slots?from=...&to=...&timezone=Asia/Kolkata	Timezone-aware inputs; only published/available slots returned.
Create booking	POST /api/bookings { therapistId, slotId }	Employee session; pseudo-ID binding; transactional conflict check; idempotency key recommended.
Therapist appointments	GET /api/therapist/appointments?from=...&to=...	Therapist role; assignment scope; therapist-safe DTO excludes corporate identity.
Wellness indicators	POST /api/wellness/indicators; GET /api/wellness/indicators	Employee can access only own wellness records; input validation and audit.
Internal repository boundary	Repository methods accept domain IDs and return typed domain objects.	Parameterized SQL; transactions; never return identity-mapping data to therapist controller.

9. Data and Security Design

- Separate EmployeeIdentity from EmployeePseudoIdentity. The mapping between corporate identity and pseudo-ID is restricted to the anonymization/identity boundary.
- Store appointment, assessment and wellness records against an internal employee key or pseudo-ID reference, not against corporate email in therapist-facing tables or API payloads.
- Enforce employee and therapist roles server-side; hiding a UI button is not authorization.

- Use TLS for network traffic, parameterized SQL for persistence, secure password/session handling, and audit records for sensitive reads and writes.
- Apply least-privilege database access: therapist read paths must not be able to query the protected identity mapping.
- Store appointment timestamps in UTC and convert to a validated IANA time zone for display; persist the selected time zone where needed for user context.
- Use database transactions and a unique constraint for appointment slot allocation to prevent concurrent duplicate bookings.
- Retain and delete wellness/counseling data according to the institution/company privacy policy; this policy needs stakeholder confirmation before production use.

10. Requirement-to-Component Traceability

Requirement	Primary components	Design response
FR-001	Anonymization; Authentication & RBAC; Repository; Database	Generate pseudo-ID; isolate identity mapping; exclude corporate identity from therapist-facing booking responses.
FR-002	API Layer; Assessment Module; Database; Web UI	Validate assessment answers, compute result, persist and show result to employee.
FR-003	Appointment Module; API Layer; Database; Web UI	Display slots, convert time zones and transactionally confirm an available slot.
FR-004	Therapist Workspace; Authentication & RBAC; Repository	Restrict to assigned therapist appointments and return pseudo-ID-only response data.
FR-005	Wellness Tracking; Database; Web UI	Save and retrieve time-ordered personal indicators.
NFR-001	Appointment Module; Database; API Layer	Canonical UTC storage, time-zone conversion and performance testing under simulated peak load.
NFR-002	Authentication & RBAC; Anonymization; Therapist Workspace; Repository; Database	Least privilege, role enforcement, isolated identity mapping, therapist-safe DTOs and audit logging.

11. Risks, Limitations and Mitigations

Risk / limitation	Impact	Mitigation
Accidental identity leakage	High confidentiality breach if therapist endpoints expose corporate identity.	Define therapist-safe DTOs; isolate identity mapping; add automated tests asserting that corporate identity fields are absent.
Concurrent slot booking	Two employees may attempt to book the same slot.	Transaction plus uniqueness constraint; return conflict response to losing request.
Time-zone ambiguity	Incorrect local display may cause missed sessions.	Store UTC, validate IANA zone names, test daylight-saving transitions and multiple office regions.
Shared database coupling	Modules can become tightly coupled through direct table access.	Use repository interfaces and explicit module APIs; prohibit direct cross-module database access.
Growth beyond one deployable unit	Single application may become difficult to scale independently.	Measure load and deployment friction; extract services only when evidence supports the cost.

12. Verification Plan

Test area	Representative verification
Pseudo-ID privacy	Create employee booking and inspect therapist response payload/database view; assert no email or employee ID is present.
Assessment flow	Submit valid and invalid answer sets; check validation, calculated result, persistence and employee-only visibility.
Booking integrity	Request available slots in multiple time zones; run concurrent booking attempts for one slot; confirm one booking only.
Therapist authorization	Authenticate as therapist A and attempt to access therapist B appointments; verify denial.
Wellness tracking	Create several indicators and retrieve history; verify ordering and employee ownership restrictions.
Security	Test unauthorized requests, role escalation, parameterized query behavior and audit logging.
Performance	Run simulated peak-load tests for slot retrieval and booking; record latency and error rate against targets agreed with stakeholders.

13. Conclusion

The modular monolithic architecture provides a practical, testable design for the current Corporate Mental Wellness & Counseling Platform while preserving clear component responsibilities. It directly supports the five functional requirements and gives particular attention to the platform's strongest constraints: employee anonymity, therapist access boundaries, appointment integrity, time-zone correctness and protection of wellness information. The microservices alternative is documented as a future scaling option, not selected without operational evidence.

Appendix A - Design Assumptions

- The implementation will expose a web-based employee and therapist client. The exact frontend framework is a project implementation decision, not a Lab 1 requirement.
- A relational database is assumed because appointment allocation, identity mapping and wellness records require durable storage and transactional constraints.
- Therapists have assigned appointment scope; the assignment-management process is outside the current Lab 1 functional requirement list.
- Assessment scoring rules and the exact wellness indicator catalogue must be agreed with stakeholders; this architecture does not invent clinical thresholds.
- Production deployment, regulatory obligations and retention periods require organizational and legal review before real employee health information is processed.

Appendix B - Prepared Diagram Files

The repository submission package includes the two exported PNG architecture alternatives and an editable diagrams.net component diagram source. The editable source corresponds to the selected modular-monolith component view.